



Request: “Find and buy a pink living room pillow.” (Missing Target Task)
[After Turn 1 search: agent finds no pink pillows exist in the environment.]

Timely success

t1	t2	t3	t4	t5	t6	t7	t8	t9	t10
ACT	ABSTAIN	—	—	—	—	—	—	—	—

Earliest possible
abstention

Delayed success

ACT	ACT	ACT	ACT	ABSTAIN	—	—	—	—	—
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Unnecessary tool calls

Failure to abstain

ACT	ACT	ACT	ACT	ACT	ACT	ACT	ACT	ACT	ACT
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Unnecessary tool calls; No abstention within 10-turn budget

ACT = tool call or environment interaction; **ABSTAIN** = stop and explain infeasibility / request clarification; **ANSWER** = task completion attempt.